

Example 4: Solve for x ,

(a) $\ln(2x^2+1) = 5 + 4\ln x$

(b) $e^{3x} > 10e^x$

(c) $\ln(2x) < 3$

Example 3: solve for x ,

(a) $\ln(3x-2) > 1$

(b) $e^{x^2} \leq e$

(c) $\ln \left| \frac{2x-1}{4x} \right| = 3$

Example 4 :

$$(a). \quad \ln(2x^2+1) = 5 + 4\ln x$$

$$\Rightarrow \ln(2x^2+1) - 4\ln x = 5$$

$$\Rightarrow \ln(2x^2+1) - \ln(x^4) = 5$$

$$\Rightarrow \ln\left(\frac{2x^2+1}{x^4}\right) = 5$$

$$\Rightarrow \frac{2x^2+1}{x^4} = e^5$$

$$\Rightarrow e^5 x^4 - 2x^2 - 1 = 0$$

$$\Rightarrow \Delta = 4 - 4x - e^5 = 4(1+e^5) > 0$$

$$\Rightarrow x = \frac{2 \pm 2\sqrt{1+e^5}}{2e^5}$$

$$\Rightarrow x_1 = \frac{1 + \sqrt{1+e^5}}{e^5} \quad \text{or} \quad x_2 = \frac{1 - \sqrt{1+e^5}}{e^5}$$

$$\boxed{\begin{array}{l} x_1 \approx 9.08 \\ x_2 \approx -9.07 \end{array}}$$

$$\Rightarrow x = \frac{1 + \sqrt{1+e^5}}{e^5} \quad \text{since } x_2 < 0$$

(observe that $\ln(2x^2+1)$ is defined $\forall x \in (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \cup (\frac{1}{\sqrt{2}}, +\infty)$
and $\ln(x)$, $\forall x \in (0, +\infty)$ so this
equation exists iff $x > \frac{1}{\sqrt{2}}$)

$$(b) e^{3x} > 10e^x$$

$$\Rightarrow \ln(e^{3x}) > \ln(10e^x)$$

since $\ln x$
is increasing
on $(0, \infty)$

$$\Rightarrow 3x > \ln(10) + \ln(e^x)$$

$$\Rightarrow 3x > \ln 10 + x$$

$$\Rightarrow x > \frac{\ln 10}{2} = \ln \sqrt{10}$$

$$(c) \ln(2x) < 3$$

$$\Rightarrow e^{\ln(2x)} < e^3$$

since e^x is increasing
on $\mathbb{R}$

$$\Rightarrow 2x < e^3$$

$$\Rightarrow x < \frac{e^3}{2}$$

$$\Rightarrow 0 < x < \frac{e^3}{2}$$

since $\ln(2x)$ is defined
 $\forall x > 0$

Example 3:

$$(a) \ln(3x-2) > 1$$

$$\Rightarrow \ln(3x-2) > \ln e$$

$$\Rightarrow e^{\ln(3x-2)} > e^{\ln e}$$

since e^x is
increasing $\forall x \in \mathbb{R}$

$$\Rightarrow 3x-2 > e$$

$$\Rightarrow x > \frac{e+2}{3}$$

$$(b) e^{x^2} \leq e'$$

$$\Rightarrow \ln(e^{x^2}) \leq \ln e'$$

since $\ln x$ is
increasing $\forall x \in (-\infty, \infty)$

$$\Rightarrow x^2 \leq 1$$

$$\Rightarrow \sqrt{x^2} = |x| \leq 1$$

$$\Rightarrow -1 \leq x \leq 1$$

$$(c) \ln \left| \frac{2x-1}{4x} \right| = 3$$

$$\Rightarrow e^{\ln \left| \frac{2x-1}{4x} \right|} = e^3$$

$$\Rightarrow \left| \frac{2x-1}{4x} \right| = e^3$$

$$\Rightarrow |2x-1| = e^3 |4x|$$

$$\Rightarrow \sqrt{(2x-1)^2} = e^3 \sqrt{(4x)^2}$$

$$\Rightarrow (2x-1)^2 = e^6 (4x)^2$$

$$\Rightarrow 4x^2 - 4x + 1 - e^6 \cdot 16x^2 = 0$$

$$\Rightarrow 4(1-4e^6)x^2 - 4x + 1 = 0$$

$$\Rightarrow \Delta = 16 - 16(1-4e^6) = 4 \times 16e^6 > 0$$

$$\Rightarrow x = \frac{4 \pm \sqrt{\Delta}}{8(1-4e^6)} = \frac{4 \pm 8e^3}{8(1-4e^6)} = \frac{1 \pm 2e^3}{2-8e^6}$$

$$\Rightarrow x = \frac{1-2e^3}{2-8e^6} \quad \text{or} \quad x = \frac{1+2e^3}{2-8e^6}$$

$$(\approx 0,012) \quad (\approx -0,012)$$

observe that $\ln \left| \frac{2x-1}{4x} \right|$ is defined

for all $x \in (-\infty, 0) \cup (0, \frac{1}{2}) \cup (\frac{1}{2}, +\infty)$